Purrfect Cut Sampler

Studio 87. Free and open source (AGPL-3.0-or-later).

Intro

Purrfect Cut is an audio editor and a sampler with FX, as a desktop app and a VST3 plugin. It is not a pitch corrector or an autotune.

Quick start: the editor app (Purrfect Cut)

- Open a file. Select a part of the waveform and cut, copy, paste, trim, silence, fade, change gain, normalise, reverse, invert or remove DC. Undo goes back 25 steps. Save as a 24 bit WAV.
- Zoom with the mouse wheel around the pointer, scroll with shift or the sideways wheel, or use the zoom in, out, fit and selection buttons.

Quick start: the sampler (Purrfect Cut Sampler, VST3 and standalone)

- Load a sample, then play it from MIDI. The sample path is saved in your project (the audio is not).
- Note: the sampler window is the plain list of controls for now. Every control is an automatable host parameter. The designed window is in the mockup in the repository and is not built yet.

Sampler

- Mode: Keys (pitch from the note) or Slices (each key plays a slice). Voices: polyphonic or mono with Glide. Root note, Tune, Fine tune and Bend.
- Sample: Region start and end, Start, Loop with Loop start and end, Reverse, Slice by (even or transients) and Slices count.
- Amp: Attack, Decay, Sustain, Release, Gain, Velocity, Pan.
- Filter: type (off, low, high, band pass), Cutoff, Resonance, Env amount, Key track.

FX chain

- Drive (Drive, Transistor, Fuzz and Mix), Squash, Duck (amount, shape, rate, tempo or free Hz, target, source shape or sidechain, restart on key), Delay (time, feedback, shift, shimmer, damping, mix), Reverb (type, decay, size, pre-delay, modulation, damping, shimmer pitch and amount, mix), Width and Output level.

Recommendations

- Use Slices mode on drum loops, and Duck with a sidechain input to pump a pad under a kick.
- Not tried on very long files or tuned by ear yet: please report problems on GitHub.

Interface tour

Every pink number marks a control that has an info bubble in the interactive mockup (docs/mockup), and the tables below carry the text of those bubbles. The pictures come from the mockup, which mirrors the plugin window; the compiled plugin can differ by a few pixels of layout.

Editor



Header bar

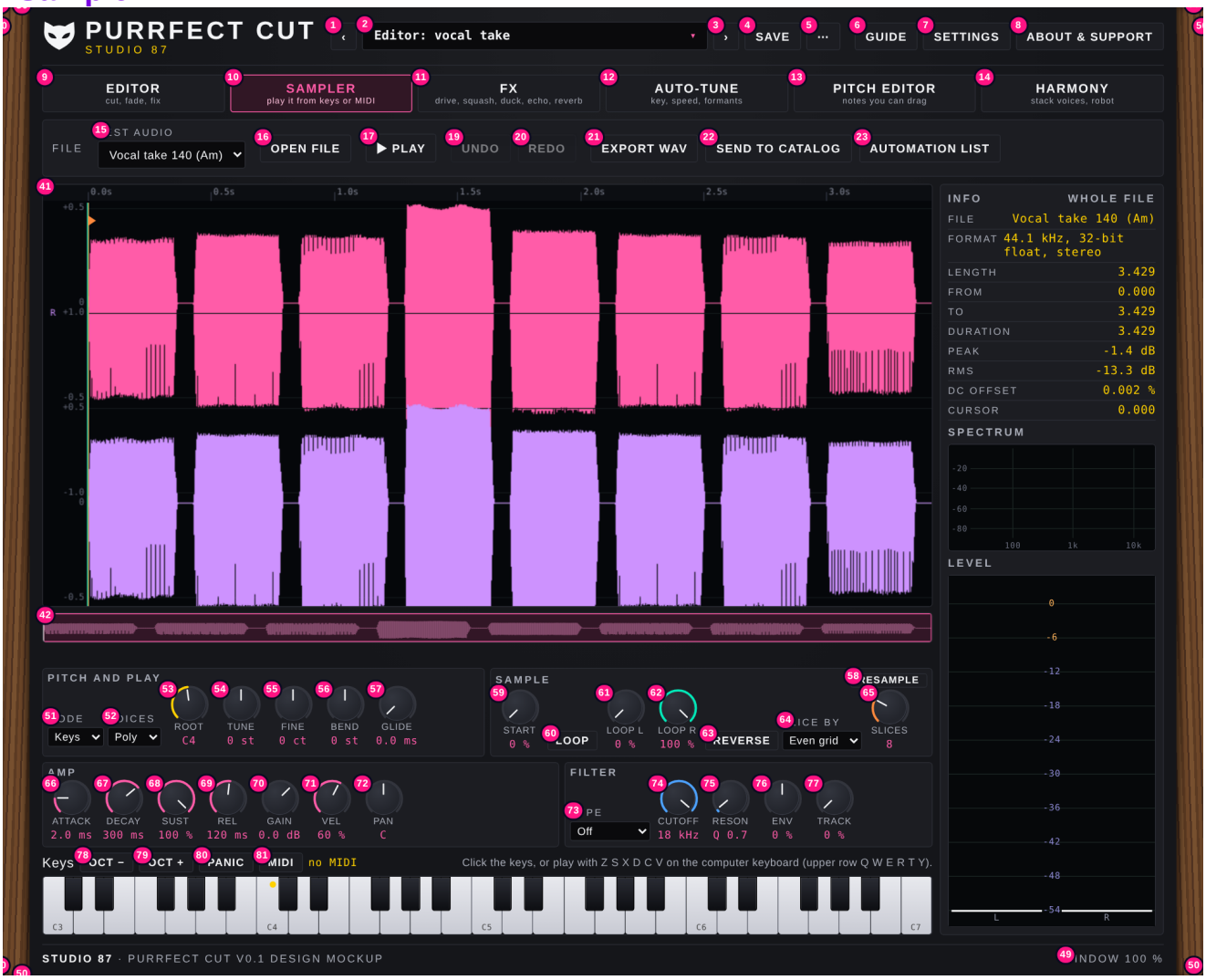
1	**Previous preset**	Previous preset.
2	**Init**	Open the preset list: factory presets and your own.
3	**Next preset**	Next preset.
4	**Save**	Save the current settings as your own preset (stays on this device).
5	**Delete, export or import your presets**	Delete, export or import your presets.
6	**Guide**	Quick guide: what each part does.
7	**Settings**	Info bubbles and eco mode.
8	**About & Support**	About Studio 87, privacy, and ways to support the project.

Window

9	**Editor**	Waveform editor: select, cut, copy, paste, trim, fades, normalize, reverse and more, with undo.
10	**Sampler**	Turn the audio (or your selection) into an instrument: play it from a keyboard or MIDI, with pitch, loop, slices, envelope and filter.
11	**FX**	Drive, Transistor and Fuzz from PurrOne, squash, duck, delay and reverb from the shimmer plugin. Works on the sampler, or on the preview, or can be applied to the audio.
12	**Auto-Tune**	Auto-Tune and pitch corrector: pulls a voice or lead onto the notes of a key, from natural to hard and robotic. Runs live in the plugin.
13	**Pitch Editor**	Melodyne style editor: the pitch of the file becomes note blobs on a piano roll. Drag notes up, down and sideways, change vibrato, drift and formants per note.
14	**Harmony**	Harmony voices (thirds, fifths, octaves that follow the key) and a robot voice locked to one note.
15	**Test audio**	Pick a test sound to edit. Reloading it throws away your edits.
16	**Open file**	Open your own audio file (wav, mp3, flac, ogg). It is edited in this browser tab only and never uploaded.
17	**Play**	Play from the cursor, or the selection. Shortcut: Space.
18	**Loop**	Loop the selection while playing. Shortcut: L (Editor).
19	**Undo**	Undo the last edit. Shortcut: Ctrl+Z.
20	**Redo**	Redo. Shortcut: Ctrl+Y.
21	**Export WAV**	Save the edited audio as a 16-bit WAV file to your computer.
22	**Send to CATalog**	Hand the edited audio to CATalog's library. Plugin and desktop app only.
23	**Automation list**	List of every control. In the plugin each one is a host parameter you can automate and MIDI map.
24	**CUT**	Remove the selection and keep a copy. Shortcut: Ctrl+X.
25	**COPY**	Copy the selection. Shortcut: Ctrl+C.
26	**PASTE**	Insert the copy at the cursor, or over the selection. Shortcut: Ctrl+V.
27	**DELETE**	Remove the selection. Shortcut: Delete.
28	**SILENCE**	Replace the selection with silence.
29	**TRIM**	Keep only the selection and remove the rest.
30	**FADE IN**	Ramp the selection (or whole file) up from silence with the chosen curve.
31	**FADE OUT**	Ramp the selection (or whole file) down to silence.
32	**NORMALIZE**	Raise or lower the selection (or whole file) so its peak hits the target level.
33	**GAIN**	Apply the gain amount below to the selection (or whole file).
34	**REVERSE**	Play the selection backwards.
35	**INVERT**	Flip the polarity of the selection.
36	**REMOVE DC**	Remove a constant offset from the selection.
37	**Zoom out**	Zoom out. Mouse wheel works too.
38	**Zoom in**	Zoom in.
39	**FIT**	Show the whole file.
40	**SELECTION**	Zoom to the selection.
41	**Waveform**	Drag to select. Shift-click extends the selection, drag its orange edges to adjust, double-click selects everything. Mouse wheel zooms, Shift+wheel scrolls.
42	**Overview**	The whole file. Drag the pink window to scroll.
43	**Gain**	Amount used by the Gain button. Double-click resets.
44	**Norm to**	Peak level the Normalize button aims for. Double-click resets.
45	**Fade curve**	Shape of the fades.
46	**Scale**	Linear shows the sample values; dB shows quiet detail better.

47	**View**	Waveform, or a spectrogram of the visible part (low frequencies at the bottom).
48	**Snap to zero**	Selection edges and the cursor jump to the nearest zero crossing, so cuts do not click.
49	**Window 100 %**	Window size. Drag any edge or corner of the plugin to resize it; the shape stays the same. Double-click here for 100 %.
50	**Resize**	Drag to resize the window. Every edge and corner works, and the shape stays the same.

Sampler



Pitch and play

51	**Mode**	Keys: the sample is pitched across the keyboard. Slices: the sample is cut into pieces and each piece sits on its own key from C4 up.
52	**Voices**	Poly: chords and overlapping notes. Mono: one note at a time, with glide between notes.
53	**Root**	The key that plays the sample at its original pitch. Double-click resets.
54	**Tune**	Transposes everything in semitones. Double-click resets.
55	**Fine**	Fine tune in cents. Double-click resets.
56	**Bend**	Pitch bend amount. A MIDI pitch wheel adds up to 2 semitones on top. Double-click resets.
57	**Glide**	Mono mode: slide time between notes. Double-click resets.

SampleResample

58	**Resample**	Plays the sample at its root note through the amp, filter and FX, and replaces the selection (or the whole file) with the result. Undo brings the old audio back.
59	**Start**	Where in the sample (or selection) the note begins. Also the orange line on the waveform. Double-click resets.
60	**Loop**	Keys mode: keeps repeating between the two green markers while the key is held. Drag the markers on the waveform.
61	**Loop L**	Loop start inside the sample. Drag the green L marker on the waveform. Double-click resets.
62	**Loop R**	Loop end inside the sample. Drag the green R marker on the waveform. Double-click resets.
63	**Reverse**	Plays the sample backwards.
64	**Slice by**	Even grid: equal pieces. Transients: cuts where the sound hits (drums, chops).
65	**Slices**	How many pieces in Slices mode. Slice 1 is C4, slice 2 is C#4, and so on. Double-click resets.

Amp

66	**Attack**	Fade in time. Double-click resets.
67	**Decay**	Time to fall from the peak to the sustain level. Double-click resets.
68	**Sust**	Level while the key is held. Double-click resets.
69	**Rel**	Fade out time after you let go. Double-click resets.
70	**Gain**	Level of the sampler. Double-click resets.
71	**Vel**	How much the key velocity changes the volume. Double-click resets.
72	**Pan**	Stereo position. Double-click resets.

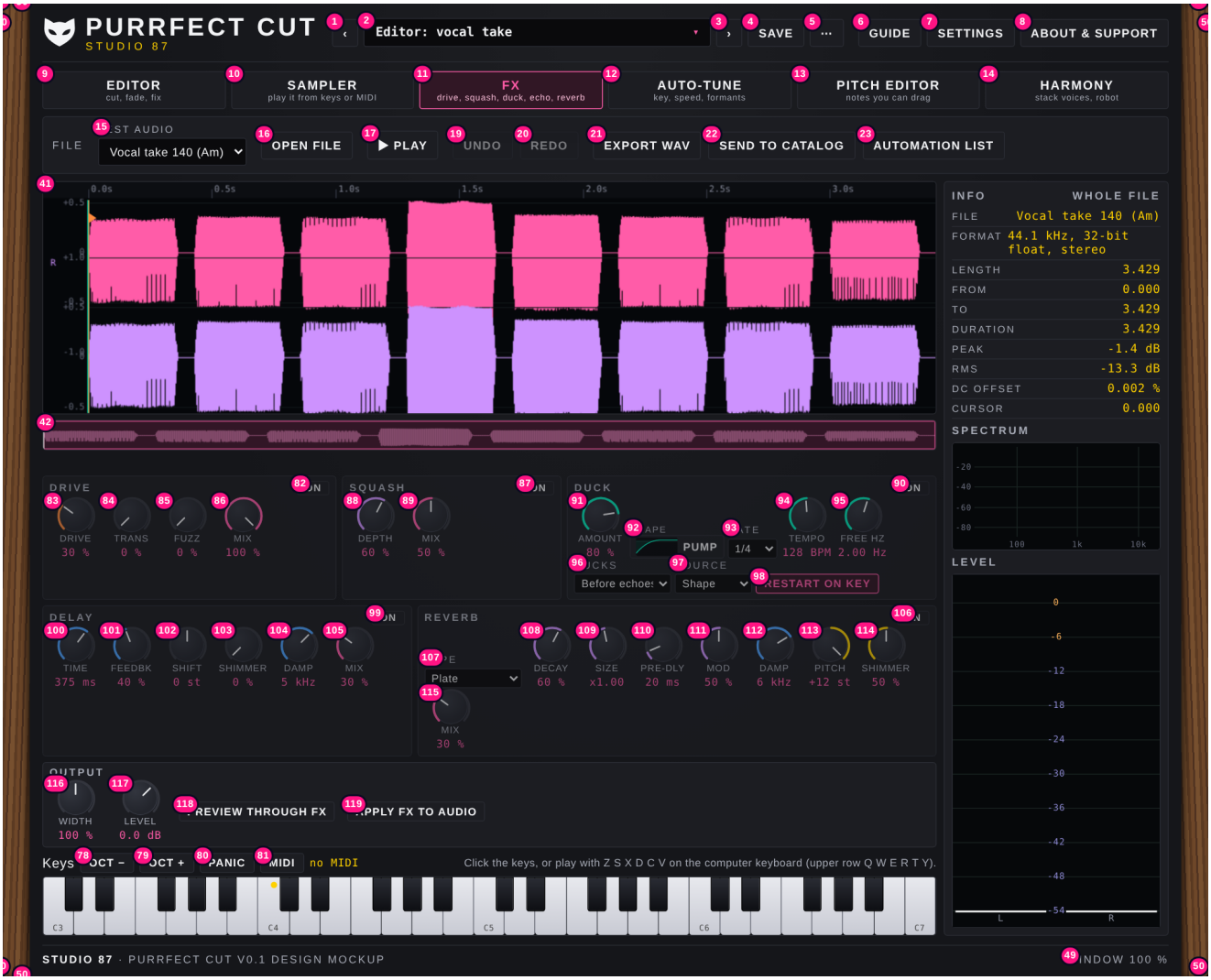
Filter

73	**Type**	Filter on every note.
74	**Cutoff**	Filter frequency. Double-click resets.
75	**Reson**	Emphasis at the cutoff. Double-click resets.
76	**Env**	How far the amp envelope moves the cutoff, up to 5 octaves. Negative sweeps down. Double-click resets.
77	**Track**	Higher keys open the filter more. Double-click resets.

Window

78	**Oct** –	Move the computer keyboard one octave down.
79	**Oct +**	Move the computer keyboard one octave up.
80	**Panic**	Stops every sounding note at once.
81	**MIDI**	Connect a MIDI keyboard (asks your browser for permission). Note on and off and pitch bend are used.

FX



DriveOn

82	**On**	Turns Drive on or off.
83	**Drive**	PurrOne's tanh drive. Smooth and round. Past 80 % a second stage stacks on. Double-click resets.
84	**Trans**	PurrOne's Neve style transistor stage: asymmetric clipping, warm even harmonics, a bit more low end. Double-click resets.
85	**Fuzz**	PurrOne's fuzz: harder and lopsided. Past 80 % it adds an octave up growl. Double-click resets.
86	**Mix**	Dry/wet of the three stages together. Double-click resets.

SquashOn

87	**On**	Turns Squash on or off.
88	**Depth**	Three band push: quiet parts up, loud parts down (OTT style). Dense, bright, loud. Double-click resets.
89	**Mix**	Dry/wet of the squash. Double-click resets.

DuckOn

90	**On**	Turns Duck on or off.
91	**Amount**	How deep the level dips. Double-click resets.
92	**Shape**	Pick the dip shape: pump like a sidechain kick, saw, sine, triangle, trance gate and more.
93	**Rate**	Speed of the shape as a note length at the tempo, or Free for a rate in Hz.
94	**Tempo**	Tempo for the synced rates. The plugin follows the host tempo. Double-click resets.
95	**Free Hz**	Speed when Rate is Free. Double-click resets.
96	**Ducks**	Before echoes: the sound dips and the delay and reverb tails keep ringing. Tails only: the dry sound stays and only the delay and reverb dip. Everything: the whole output dips.
97	**Source**	Shape: the dip follows the chosen shape. Sidechain: the dip follows the loudness of the audio you are previewing in the Editor (the plugin uses its sidechain input). Turn off 'Preview through FX' to use it.
98	**Restart on key**	The shape starts from the top every time you press a key.

DelayOn

99	**On**	Turns Delay on or off.
100	**Time**	Time between repeats. Double-click resets.
101	**Feedbk**	How much of each repeat goes back in. Double-click resets.
102	**Shift**	Pitch change at every repeat (shimmer delay). Double-click resets.
103	**Shimmer**	How much of each repeat is shifted. 0 is a plain delay. Double-click resets.
104	**Damp**	Darkens every repeat. Double-click resets.
105	**Mix**	Dry/wet of the delay. Double-click resets.

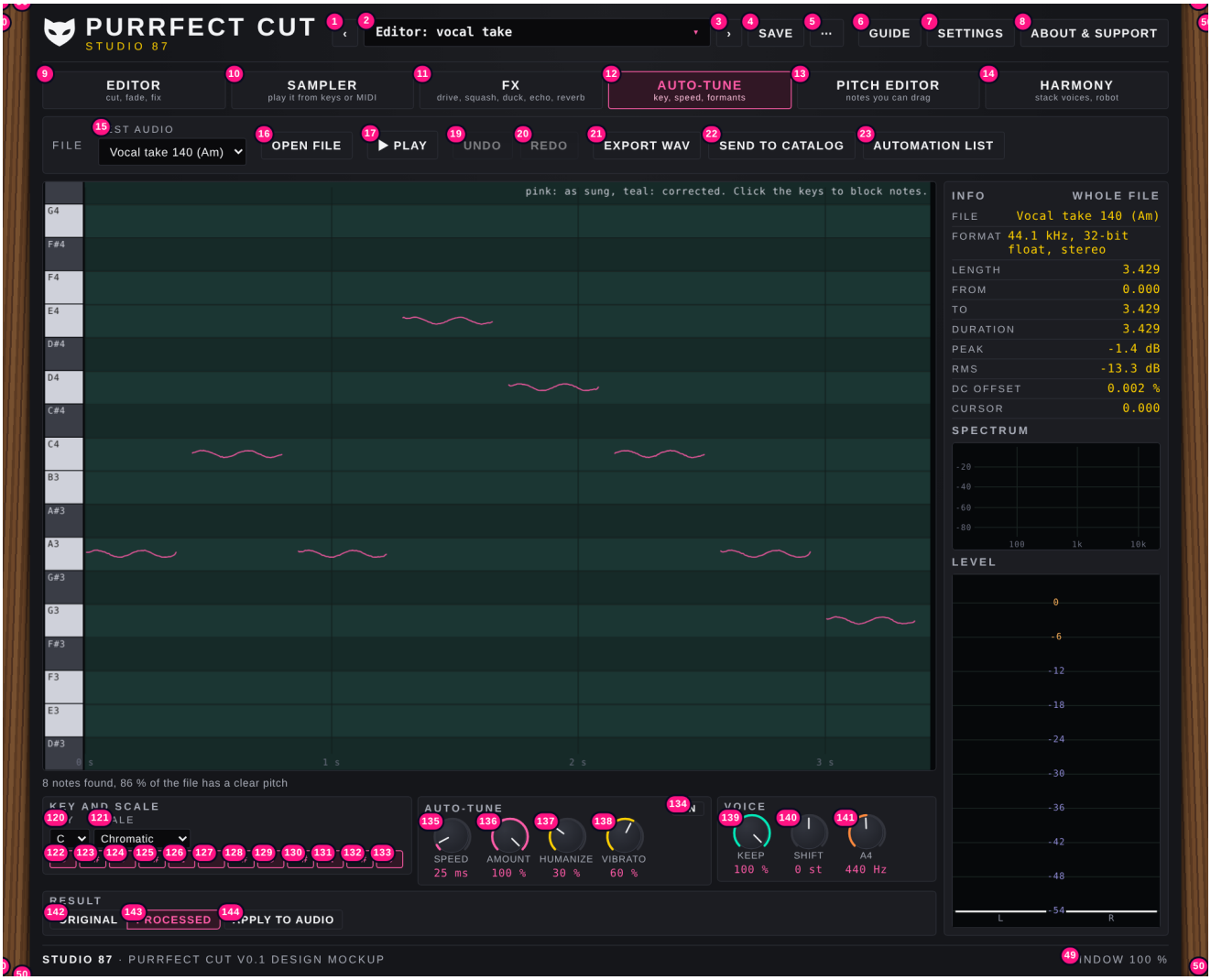
ReverbOn

106	**On**	Turns Reverb on or off.
107	**Type**	Plate: bright, short to medium. Hall: bigger and smoother. Shimmer types add a pitch shifted copy into the tail. Velvet: two halls blended.
108	**Decay**	Length of the tail. Double-click resets.
109	**Size**	Small room or big room. Double-click resets.
110	**Pre-dly**	Gap before the tail starts. Double-click resets.
111	**Mod**	Slow movement in the tail. Double-click resets.
112	**Damp**	Highs die away faster when lower. Double-click resets.
113	**Pitch**	Pitch of the shimmer copy (shimmer types only). Double-click resets.
114	**Shimmer**	Amount of the shimmer copy. Double-click resets.
115	**Mix**	Dry/wet of the reverb. Double-click resets.

Output

116	**Width**	Stereo width. 0 is mono, 200 % extra wide. Double-click resets.
117	**Level**	Output level of the sampler and FX. Double-click resets.
118	**Preview through FX**	On: the Editor preview (Play) also goes through the FX chain. Off: the preview stays clean and only the sampler gets the FX.
119	**Apply FX to audio**	Runs the selection (or the whole file) through the FX chain and writes the result into the audio, including the reverb and delay tail. Undo brings the old audio back.

Auto-Tune



Key and scale

120	**Key**	The key. Notes outside the scale are pulled to the nearest note inside it.
121	**Scale**	The scale. Chromatic lets every note through (the pitch is rounded to the nearest semitone).
122	**Allow or block C on top of the scale**	Allow or block C on top of the scale. You can also click the keys on the left of the graph.
123	**Allow or block C# on top of the scale**	Allow or block C# on top of the scale. You can also click the keys on the left of the graph.
124	**Allow or block D on top of the scale**	Allow or block D on top of the scale. You can also click the keys on the left of the graph.
125	**Allow or block D# on top of the scale**	Allow or block D# on top of the scale. You can also click the keys on the left of the graph.
126	**Allow or block E on top of the scale**	Allow or block E on top of the scale. You can also click the keys on the left of the graph.
127	**Allow or block F on top of the scale**	Allow or block F on top of the scale. You can also click the keys on the left of the graph.
128	**Allow or block F# on top of the scale**	Allow or block F# on top of the scale. You can also click the keys on the left of the graph.
129	**Allow or block G on top of the scale**	Allow or block G on top of the scale. You can also click the keys on the left of the graph.
130	**Allow or block G# on top of the scale**	Allow or block G# on top of the scale. You can also click the keys on the left of the graph.

131	**Allow or block A on top of the scale**	Allow or block A on top of the scale. You can also click the keys on the left of the graph.
132	**Allow or block A# on top of the scale**	Allow or block A# on top of the scale. You can also click the keys on the left of the graph.
133	**Allow or block B on top of the scale**	Allow or block B on top of the scale. You can also click the keys on the left of the graph.

Auto-TuneOn

134	**On**	Turns Auto-Tune on or off.
135	**Speed**	Retune speed. 0 snaps at once (the hard, robotic effect). 20 to 60 ms is a clean modern vocal, 100 ms and up keeps it natural. Double-click resets.
136	**Amount**	How far the pitch is pulled towards the note. 100 % lands on it, 50 % lands half way. Double-click resets.
137	**Humanize**	Slows the correction down the longer a note is held, so long notes can drift a little like a real singer. Double-click resets.
138	**Vibrato**	How much of the singer's own vibrato is kept. 0 flattens it (robotic), 100 % keeps all of it. Double-click resets.

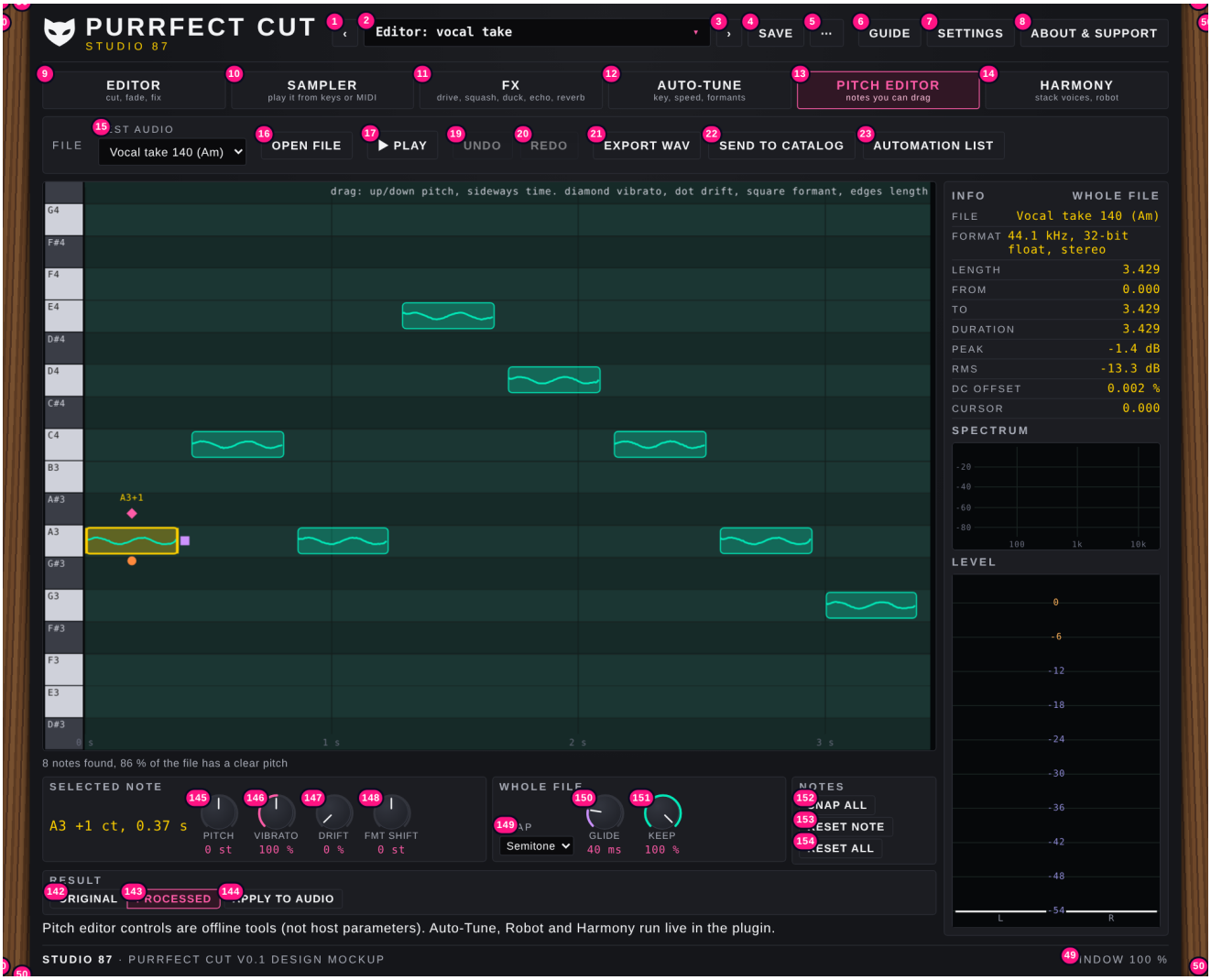
Voice

139	**Keep**	Formant keep. 100 % keeps the vowel colour when the pitch moves (the voice keeps its character). 0 resamples, so the voice gets smaller or bigger with the pitch. Double-click resets.
140	**Shift**	Formant shift. Moves the formants on their own: up for a smaller, younger voice, down for a bigger one. The pitch stays. Double-click resets.
141	**A4**	The reference pitch for A4. 440 Hz is standard. Double-click resets.

Result

142	**Original**	Plays the file as it is, to compare.
143	**Processed**	Plays the file with the pitch changes. Space plays it from the cursor.
144	**Apply to audio**	Writes the processed pitch into the file (all of it). Undo brings the old audio back.

Pitch Editor



Selected note

145	**Pitch**	Moves the note up or down. You can also drag the note in the graph. Fine: hold Alt while dragging. Double-click resets.
146	**Vibrato**	Scales the note's own vibrato: 0 flat, 100 % as sung, 200 % twice as wide. The pink diamond above the note does the same. Double-click resets.
147	**Drift**	Straightens slow drift away from the note's centre (a note that sags or climbs). The orange dot under the note does the same. Double-click resets.
148	**Fmt shift**	Formant shift for this note only. The purple square on the right of the note does the same. Double-click resets.

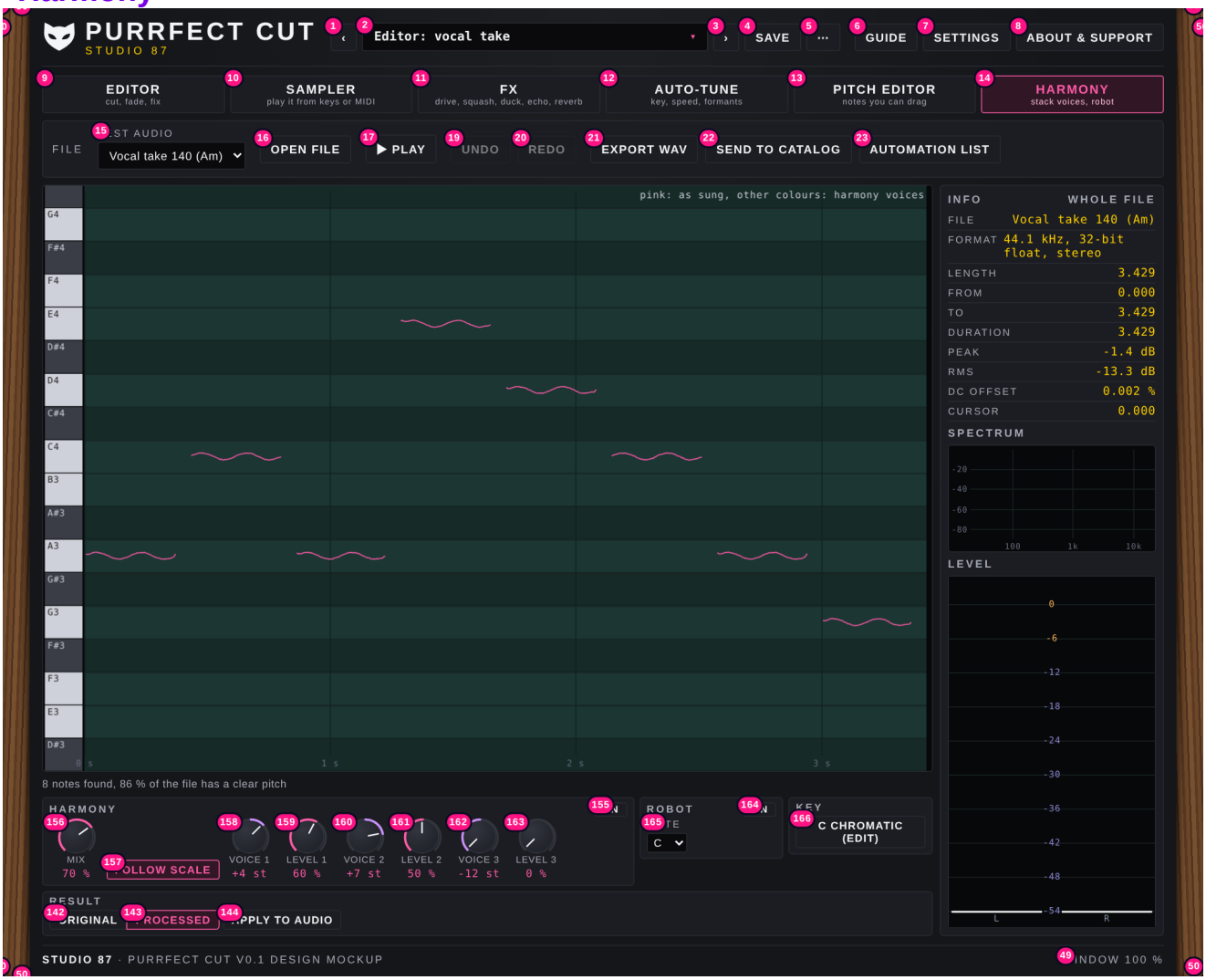
Whole file

149	**Snap**	While you drag a note: Off is free, Semitone lands on whole notes, Scale lands on notes of the key set on the Auto-Tune page.
150	**Glide**	How smoothly the pitch moves from one edited note to the next. Double-click resets.
151	**Keep**	100 % keeps the vowel colour when notes move. 0 lets the voice change size with the pitch. Double-click resets.

Notes

152	**Snap all**	Moves every note to the nearest note of the scale.
153	**Reset note**	Puts the selected note back as it was sung.
154	**Reset all**	Puts every note back as it was sung.

Harmony



HarmonyOn

155	**On**	Turns Harmony on or off.
156	**Mix**	Level of all harmony voices together, against the main voice. Double-click resets.
157	**Follow scale**	On: each voice is pulled into the key, so a third stays a third of the scale (major or minor as the key needs). Off: exact intervals.
158	**Voice 1**	Interval of voice 1 from the singer, in semitones (4 is a third, 7 a fifth, 12 an octave). Double-click resets.
159	**Level 1**	Level of voice 1. 0 switches it off. Double-click resets.
160	**Voice 2**	Interval of voice 2 from the singer, in semitones (4 is a third, 7 a fifth, 12 an octave). Double-click resets.
161	**Level 2**	Level of voice 2. 0 switches it off. Double-click resets.
162	**Voice 3**	Interval of voice 3 from the singer, in semitones (4 is a third, 7 a fifth, 12 an octave). Double-click resets.
163	**Level 3**	Level of voice 3. 0 switches it off. Double-click resets.

RobotOn

164	**On**	Turns Robot on or off.
165	**Note**	Robot voice: the whole performance is forced onto this one note (in the nearest octave). Needs Auto-Tune on. Set the speed to 0 for a hard robot.

Key

166	**Key and scale**	Harmony and Robot use the key, scale and note switches of the Auto-Tune page. Click to go there.